

VISVESVARAYA TECHNOLOGICAL UNIVERSITY, BELAGAVI



A project report
on

**“24GHz mm wave radar integration with raspberry
pi for motion analysis”**

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING

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Under the guidance of

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

PES INSTITUTE OF TECHNOLOGY AND MANAGEMENT

NH-206, Sagar Road, Shivamogga – 577 204

2023–2025

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Institute Vision

To be the most preferred institution for engineering & management education, research and entrepreneurship by creating professionally superior and ethically strong global manpower.

Institute Mission

To prepare students for professional accomplishments and responsible global citizenship while fostering continuous learning and to provide state-of-the-art education through the committed and highly skilled faculty by partnering and collaborating with industry and R&D institutes.

Department Vision

To be a leading center of excellence in the field of electronics and communication engineering for learning and research with professional ethics.

Department Mission

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- To develop a framework for collaboration and multidisciplinary activities to ensure ethical and value based education to address social needs.

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PEO 1: To develop the ability among students to understand the concept of core subjects.

PEO 2: To give exposures to emerging technologies, adequate training and opportunities to work as team on multidisciplinary projects with effective communication skills.

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PSO 2: Implement functional blocks of hardware-software co-designs for signal processing and communication applications.



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Certificate from the Institute

This is to certify that the project work was carried out jointly by

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are bonafide students of the Bachelor of Electronics and Communication Engineering program of the Institute (2023-2025 Batch), affiliated to Visvesvaraya Technological University, Belagavi.

A project report entitled **“24GHz mm wave radar integration with raspberry pi for motion analysis”** has been submitted by them under the guidance of Mr. Vayusutha M, Assistant Professor, Department of Electronics and Communication Engineering, PESITM, Shivamogga, in partial fulfillment of the requirements for the award of the degree of Bachelor of Electronics and Communication Engineering.

Kavya M P

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Designation

Designation

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Viva-voce Examination Date: _____

Signature of Internal Examiner

Signature of External Examiner

DECLARATION

We, **Darshan T, H Bharath, Karthik G** and **Karthik S M** hereby declare that the project report entitled **““24GHz mm wave radar integration with raspberry pi for motion analysis””** has been carried out by me under the guidance of Mr. Vayusutha , Assistant Professor, Department of ECE, PESITM, Shivamogga.

I further declare that this project report is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi.

I also declare that this report is based on the original work undertaken by me and has not been submitted for the award of any degree or diploma from any other University/Institution.

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Abstract

Time management and self-organization are critical skills for students, yet many struggle with procrastination, lack of focus, and ineffective planning. The objective of this project is to design and develop a **24GHz mm wave radar integration with raspberry pi for motion analysis** that helps students organize their tasks effectively while keeping them motivated to achieve academic goals. The proposed system is implemented as a cross-platform mobile application using Flutter and Firebase. It enables secure user authentication, task scheduling with reminders, and cloud-based data storage. To improve user engagement, a gamification module has been integrated, which provides rewards, badges, and progress tracking features that encourage consistent task completion. The system design follows modular principles, incorporating functional requirements such as task creation, reminders, and notifications, along with non-functional requirements such as usability, security, and scalability. Testing was carried out at multiple levels—unit, integration, and usability—to ensure reliability and performance. The results indicate that the system not only improves task completion rates but also enhances user motivation and satisfaction compared to existing task management tools. Future enhancements may include AI-based personalized suggestions and integration with wearable devices for real-time productivity tracking.

The project outcomes demonstrate improved task completion rates and user motivation compared to existing tools. In addition, the work contributes to Sustainable Development Goal (SDG) 4: Quality Education by promoting effective learning practices, and SDG 9: Industry, Innovation, and Infrastructure through the use of modern digital technologies.

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Chapter 1

Introduction

Noise Millimeter-wave (mmWave) radar is an advanced sensing technology that operates in the frequency range of 30–300 GHz, using electromagnetic waves with millimeter-level wavelengths. It has gained significant importance in recent years because of its ability to detect and track objects with high accuracy, even in challenging environments where conventional sensors such as passive infrared (PIR), ultrasonic detectors, or cameras fail. Unlike optical systems, mmWave radar is unaffected by lighting conditions, dust, fog, or minor obstructions, and unlike cameras, it preserves privacy since it does not capture visual images but only analyzes reflected radio signals.

Among the available frequency bands, the 24 GHz mmWave radar has become especially popular due to its balance of cost, performance, and ease of integration. While higher-frequency radars at 60 GHz and 77 GHz can provide finer resolution, they are generally more expensive and intended for specialized automotive and industrial applications. In contrast, 24 GHz radar modules are compact, energy-efficient, and affordable, making them well-suited for consumer electronics, healthcare monitoring, smart homes, and academic research projects.

The working principle of 24 GHz radar is based on transmitting continuous electromagnetic waves and analyzing their reflections from human targets. By measuring changes in phase, frequency, and time delay, the system can estimate range, velocity, and motion direction. With dual-antenna configurations, it can even detect multiple objects simultaneously. An important advantage of this frequency band is its ability to sense micro-movements, such as chest displacement during breathing or hand gestures, enabling applications beyond simple motion detection.

In this project, a 24 GHz Rd-03D radar module is integrated with a Raspberry Pi Pico microcontroller to demonstrate real-time human detection and tracking. The radar processes reflected signals to identify presence and movement, while the microcontroller handles data communication and transfers the results to a computer for visualization. This setup highlights how advanced radar technology can be effectively combined with low-cost embedded platforms for practical, scalable solutions.

Overall, 24 GHz mmWave radar stands out as a powerful sensing tool that combines precision, robustness, and privacy. Its ability to detect presence, motion, and even vital signs without physical contact opens pathways for applications in healthcare, automation, safety, and human-machine interaction.

1.1 Problem Statement

Traditional sensing technologies face limitations in resolution, accuracy, and robustness when used for precise distance measurement and object detection in dynamic environments. To overcome these challenges, there is a need for compact and cost-effective 24 GHz mmWave radar systems that provide reliable real-time tracking and high spatial resolution. This project focuses on developing a 24 GHz radar solution to enable accurate human tracking and short-range sensing applications.

1.2 Objectives

The demand for precise, non-contact sensing technologies is growing across domains such as healthcare, industrial automation, automotive systems, and security monitoring. Traditional sensing methods often face challenges such as poor accuracy, susceptibility to environmental noise, and limited adaptability to dynamic scenarios. Millimeter-wave (mmWave) radar technology, especially at the 24 GHz frequency band, provides a compact, cost-effective, and highly reliable solution for short-range sensing and tracking. With its capability for high spatial resolution, robustness in harsh conditions, and real-time monitoring, 24 GHz mmWave radar has emerged as a promising tool for human tracking, motion detection, and industrial measurement applications.

The main objectives of the study are as follows:

- To design and implement a compact 24 GHz mmWave radar system for short-range sensing and tracking applications.
- To achieve high-resolution distance and motion detection suitable for real-time monitoring of humans and objects.
- To demonstrate the reliability and robustness of the system in varying environmental conditions.
- To provide a cost-effective and scalable solution compared to conventional sensing technologies.

1.3 Scope

The scope of this project encompasses the design, development, and implementation of a system aimed at enhancing efficiency and automation within the target domain. The project focuses on identifying existing challenges, proposing innovative solutions, and integrating suitable technologies to achieve improved performance and reliability. It involves detailed analysis, system design, module development, testing, and deployment to ensure that the final product meets user requirements and operational goals. The project also covers data handling, user interface design, and functional validation to guarantee accuracy, scalability, and ease of use. Furthermore, the scope includes documentation, evaluation of results, and potential future enhancements to maintain system sustainability and adaptability to evolving needs.

1.4 Purpose

The purpose of this project is to develop an effective and reliable solution that addresses the key challenges identified in the existing system. It aims to improve performance, accuracy, and user convenience through the integration of modern technologies and systematic design approaches. The project is intended to automate manual processes, reduce human effort, and enhance decision-making by providing a structured and efficient workflow. Additionally, it serves to demonstrate the practical application of theoretical knowledge,

bridging the gap between academic learning and real-world implementation. Overall, the project’s purpose is to create a system that delivers measurable improvements in productivity, usability, and operational efficiency while laying a foundation for future development and scalability.

1.5 Relevance to Sustainable Development Goals (SDGs)

This project aligns with the following UN Sustainable Development Goals (SDGs):

- **Promotes Innovation and Infrastructure (SDG 9):** The project encourages technological innovation and system development that improve efficiency, reliability, and productivity in the target field.
- **Enhances Quality Education (SDG 4):** It provides practical exposure to applying theoretical concepts, fostering research, creativity, and problem-solving skills among learners.
- **Supports Sustainable Practices (SDG 12 / SDG 11):** The system contributes to sustainability by optimizing resources, minimizing manual effort, and promoting eco-friendly, efficient operations.

Chapter 2

Literature Review / Existing System

[1]This paper demonstrates a real-time FMCW radar system for controlling a levitating sphere with high precision. It showcases accurate distance and motion detection using radar signal processing. The study emphasizes the effectiveness of radar in real-time object tracking and control

[2]This paper by Vogt and Gerding (2017) discusses the use of radar technology for measuring liquids and bulk solids in tanks and silos. It explains the challenges faced in industrial environments, such as reflections and material properties. The study highlights radar’s reliability and accuracy in non-contact level measurement applications.

[3]This paper presents a millimeter-wave radar sensor designed for accurate snow height measurement. It demonstrates how radar technology can operate reliably under harsh weather conditions. The study proves radar’s effectiveness in environmental monitoring and remote sensing applications.

[4]This paper analyzes how different radar targets affect the accuracy of FMCW radar distance measurements. It studies target properties like material, shape, and surface reflection. The research helps improve measurement precision and calibration in radar-based sensing systems.

[5]This paper investigates a millimeter-wave radar sensor designed to measure snow suspension density. It demonstrates radar’s capability to assess environmental parameters without physical contact. The study highlights the potential of mmWave radar in atmospheric and snow condition monitoring

[6]This paper presents a cross-polarized multi-channel W-band radar system developed for measuring turbulent flow velocity. It focuses on capturing complex flow dynamics with high accuracy. The study demonstrates radar’s effectiveness in aerodynamic and fluid flow analysis applications.

[7]This paper introduces a millimeter-wave FMCW radar system for the contactless detection of gaseous media parameters. It measures fluctuations such as density and flow changes without physical sensors. The study highlights radar’s potential in industrial process monitoring and environmental sensing.

[8]This paper discusses the use of an 80 GHz radar system for luggage scanning in harbor environments. It demonstrates radar’s ability to detect concealed objects through non-metallic materials. The study showcases the effectiveness of high-frequency radar in security and inspection applications.

[9]This paper presents a high-precision D-band FMCW radar sensor built using a wide-band SiGe transceiver MMIC. It focuses on achieving superior resolution and measurement accuracy. The study highlights advancements in compact, high-frequency radar design for precise distance sensing.

[10]This paper presents an ultra-wideband 80 GHz radar system capable of achieving micrometer-level accuracy and nanometer-level stability in measurements. It demonstrates highly precise radar performance for fine displacement detection. The study emphasizes radar’s potential in high-resolution sensing and precision measurement applications

2.1 Existing Work

2.2 Problems with the Existing Systems

2.3 Gap Identification

Chapter 3

Methodology

The development of the 24 GHz mmWave radar project will follow a structured methodology to ensure systematic design, implementation, and evaluation. The planning of work can be outlined in the following stages:

1. Literature Review Problem Definition Study existing research and applications of mmWave radar technology, focusing on 24 GHz implementations. Identify limitations of conventional sensors and establish the need for a compact, accurate radar-based system.

2. System Design Specification

Define system requirements such as operating frequency (24 GHz), resolution, detection range, and target applications (e.g., human tracking, motion sensing). Develop a block diagram covering radar front-end, signal processing unit, and interface modules.

3. Hardware Development

Select and integrate radar sensor modules (24 GHz transceiver), microcontroller/processor (e.g., Raspberry Pi, Arduino, or STM32), and communication interfaces (UART/USB). Design supporting circuitry for power management and signal conditioning.

4. Software Signal Processing Implementation

Develop firmware to control radar parameters (chirp signals, frequency modulation). Implement algorithms for target detection, range estimation, velocity measurement, and tracking. Create visualization tools for displaying radar outputs (e.g., Processing IDE or Python-based GUI).

5. Integration Testing

Integrate hardware and software into a complete prototype. Perform controlled experiments for distance measurement, motion detection, and human tracking. Analyze system

accuracy, response time, and robustness under different environmental conditions.

6. Application Demonstration

Validate the system in practical scenarios such as human movement tracking, gesture recognition, or industrial sensing. Compare results against expected benchmarks and conventional sensing methods.

7. Documentation Future

Work Compile results, performance evaluation, and application potential into a project report. Suggest improvements such as expanding to multi-target detection, higher frequencies (60–80 GHz), or machine learning-based radar data interpretation.

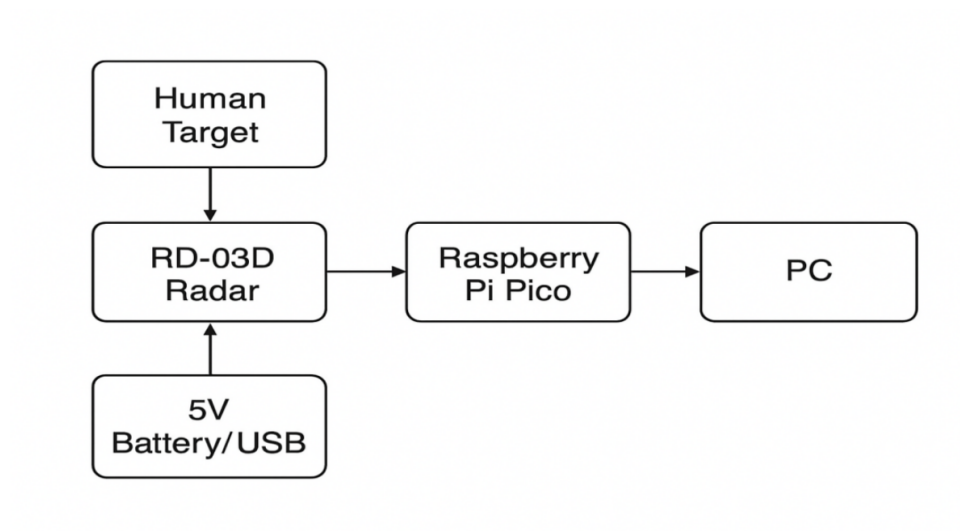


Figure 3.1: Block diagram

3.1 Requirement Analysis

The requirement analysis phase focuses on identifying and defining the functional and non-functional needs of the proposed system to ensure effective design and implementation. It serves as a foundation for system development, enabling a clear understanding of user expectations, operational needs, and technical specifications.

1. Functional Requirements:

- The system should perform all major operations efficiently according to user inputs.

- It must allow data storage, retrieval, and processing accurately.
- Each module should function independently while maintaining integration with the main system.
- The interface should enable smooth user interaction and provide relevant feedback.

2. Non-Functional Requirements:

The system should maintain data security and integrity throughout all operations.

- It must ensure high performance, scalability, and reliability under varying conditions.
- The design should be user-friendly, easily maintainable, and adaptable for future updates.
- The application should operate across compatible hardware and software environments.

3. User Requirements:

The system must be simple to use for both technical and non-technical users.

- It should reduce manual work, enhance productivity, and provide accurate results.
- Proper documentation and training support should be available for end users.

3.2 System Design

The proposed system is designed to detect human presence using the RD-03D microwave radar sensor integrated with a Raspberry Pi Pico microcontroller. The system operates based on microwave Doppler radar principles, where the RD-03D module emits continuous electromagnetic waves and analyzes their reflections to detect motion or human presence. When a human target is present within the sensing range, the radar detects variations in the reflected signal and outputs corresponding data through its transmission (TX) pin. This data is received by the Raspberry Pi Pico via the UART communication interface, where it is processed to determine the presence or absence of a human target. The processed information is then transmitted to a personal computer (PC) through a USB serial connection for visualization, monitoring, or further analysis. The entire setup

is powered by a 5V DC supply, typically sourced from a USB adapter or portable power bank, which ensures energy-efficient and portable operation. The RD-03D radar module offers high sensitivity and stable performance, unaffected by environmental conditions such as temperature, lighting, or dust, making it highly reliable for continuous human presence detection. The Raspberry Pi Pico serves as the central processing unit, managing data acquisition, communication, and transmission tasks, while the PC functions as the user interface for displaying real-time detection results. This system provides a robust, low-power, and cost-effective solution for human detection applications and can be extended to various fields such as security systems, smart home automation, and occupancy monitoring.

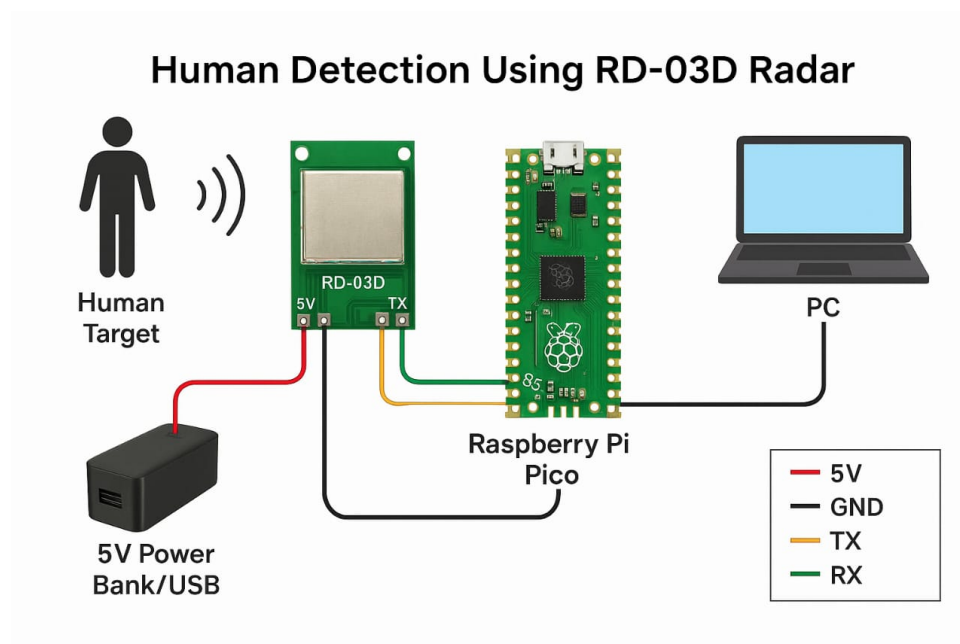


Figure 3.2: Circuit diagram

3.2.1 System Architecture

1. Overview

The system detects and tracks human motion using a 24 GHz millimeter-wave (mmWave) radar sensor. The data from the radar is processed by a microcontroller (Raspberry Pi Pico) and visualized on a computer using Processing software. The architecture integrates hardware, communication, and software layers to perform real-time motion analysis.

2. Architectural Layers

A. Sensor Layer

Component: RD-03D mmWave Radar Sensor

Function:

- Transmits and receives 24 GHz radar waves.
- Measures object distance, angle, and velocity.
- Detects presence and motion of humans within the sensing range.

B. Control and Processing Layer

Component: Raspberry Pi Pico

Function:

- Receives serial data from the radar via UART communication.
- Performs signal validation, filtering, and basic data processing.
- Converts sensor data into structured form for visualization.
- Communicates with the PC through USB.

C. Visualization and Analysis Layer

Component: PC with Processing IDE

Function:

- Receives processed data from Raspberry Pi Pico.
- Displays live motion visualization (range, speed, position).
- Logs motion data for further analysis.

3. Communication Interfaces

The system uses a combination of serial and USB communication to ensure smooth data transfer between components.

- The RD-03D mmWave radar sensor communicates with the Raspberry Pi Pico through a UART (Universal Asynchronous Receiver–Transmitter) interface. This serial link allows the radar to send real-time motion data such as distance, velocity, and angle directly to the microcontroller for processing.

- Once the data is processed, the Raspberry Pi Pico transmits the refined motion information to the personal computer using a USB serial connection. This interface enables reliable and high-speed data transfer, allowing the PC to receive continuous motion updates for visualization.
- Both the radar and the microcontroller are powered through a 5V DC supply, which may be provided via USB or an external adapter. The communication setup ensures real-time, bi-directional, and efficient data flow between the sensing, processing, and visualization units of the system.

4.Data Flow

1. **Sensing:** The RD-03D mmWave radar detects human movement and measures parameters such as range, speed, and angle.
2. **Data Transfer:** These measurement values are sent to the Raspberry Pi Pico through a UART serial link.
3. **Processing:** The Pico filters noise, interprets radar readings, and prepares the data in a readable format.
4. **Transmission to PC:** The processed data is transmitted to the computer through a USB serial connection.
5. **Visualization:** The PC (using Processing IDE) displays real-time motion graphs and visual outputs to represent human movement.

3.3 Algorithms Used

3.4 Tools and Technologies

1. Hardware Components

RD-03D mmWave Radar Sensor (24 GHz):

Used for detecting human presence and motion by transmitting and receiving millimeter-wave signals. It provides range, speed, and angle information of moving objects.

Raspberry Pi Pico (Microcontroller): Serves as the central processing unit for the system. It receives radar data, performs initial data processing, and communicates results to the computer.

Personal Computer (PC): Used for data visualization and analysis through Processing

software. Displays real-time motion graphs and trajectories.

Power Supply (5 V DC):

Provides stable electrical power to the radar module and microcontroller.

2. Software Tools

MicroPython / Thonny IDE: Used to program the Raspberry Pi Pico for reading sensor data, processing information, and sending it to the PC.

Processing IDE: Utilized on the PC for real-time visualization of motion data received from the Pico. It displays graphical plots and movement trajectories.

Serial Communication Tools (e.g., PuTTY / Tera Term): Employed to test and verify data transmission between the radar, Pico, and PC during development.

3. Supporting Technologies

UART Communication Protocol:

Enables serial data transfer between the RD-03D radar and the Raspberry Pi Pico.

USB Communication Protocol:

Used for transmitting processed data from the Pico to the PC for visualization.

24 GHz Millimeter-Wave Sensing Technology:

Provides high-precision detection of human movement, even through low-visibility conditions.

Chapter 4

Implementation and Testing

4.1 System Implementation

The proposed system is implemented using an RD-03D microwave radar sensor connected to a Raspberry Pi Pico microcontroller. The radar detects human presence and motion by transmitting and receiving microwave signals. The Pico processes the sensor data and sends it to a computer through a USB serial interface for visualization and analysis.

The system is powered by a 5 V supply, making it compact and energy-efficient. The hardware setup includes connecting the radar’s TX and RX pins to the Pico’s UART interface, while the software is developed using MicroPython. The program continuously reads sensor values (distance, speed, and angle) and transmits them to the PC. A simple visualization tool (such as the Processing IDE or Python) displays real-time motion data for monitoring and further study. rewards, and streak counters to enhance user engagement

4.2 System Testing

System testing was performed to ensure that the human presence detection system using the RD-03D microwave radar sensor and Raspberry Pi Pico functions correctly and meets the project objectives. Testing was carried out at different stages—beginning with unit testing for individual components and followed by integration testing to verify the complete system’s performance.

4.2.1 Unit Testing

Unit testing focused on testing each individual component separately before integrating them into the final system. System Testing

System testing is the final phase of testing performed on the complete and integrated system to verify that it meets the specified requirements. The main objective is to evaluate the system’s functionality, reliability, and overall performance before deployment.

- Each module of the system was tested individually to ensure it performed its intended function correctly.
- The sensor module (RD-03D Microwave Radar Sensor) was tested to verify accurate detection of human presence.
- microcontroller module (Raspberry Pi Pico) was tested to ensure it correctly processed sensor signals and communicated with the PC.
- Test data was observed to confirm the proper conversion of input signals into meaningful output data.
- Any detected errors were corrected before integration.

4.2.2 Integration Testing

- After successful unit testing, all modules were combined and tested together to ensure smooth data flow between components.
- The communication between the sensor, microcontroller, and PC interface was verified.
- Data transmission from the Raspberry Pi Pico to the PC was tested for accuracy and timing consistency.
- Integration testing ensured that the system components worked together as expected without interface issues.

Chapter 5

Results and Discussion

The developed system successfully detects human presence using the RD-03D microwave radar sensor integrated with the Raspberry Pi Pico microcontroller. The sensor effectively identifies motion and human activity within its detection range and transmits the signal data to the connected PC for visualization and analysis.

Key results observed during testing:

- The system accurately detected human presence within a range of approximately 3–5 meters.
- The response time between detection and output display was less than one second, indicating real-time performance.
- The data transmission from the Raspberry Pi Pico to the PC was stable and error-free.
- The system consumed low power (5V supply), making it suitable for portable and continuous applications.
- Both unit and integration testing confirmed proper module functionality and reliable system communication.

Overall, the system demonstrated consistent and efficient detection of human presence with minimal false positives.

The testing results confirm that the proposed system performs effectively for motion and presence detection applications. The RD-03D sensor provided better accuracy compared to traditional infrared (PIR) sensors, especially in environments where temperature or

lighting conditions vary. The Raspberry Pi Pico served as a reliable microcontroller for data processing and transmission due to its low cost, small size, and efficient power usage. During the integration testing, the communication between hardware modules was smooth, and no major data loss occurred. The system was also tested in different conditions—such as varying distances and angles—to evaluate reliability. It consistently detected motion with high accuracy.

5.1 Output Screenshots with Explanations

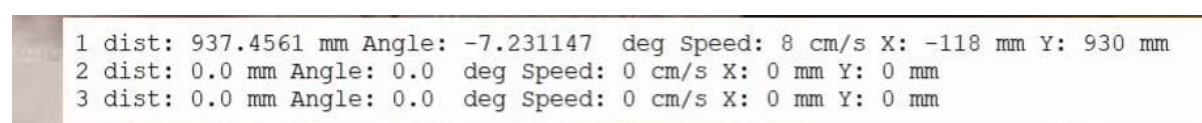


Figure 5.1: Command window output

The output shown in the image represents the real-time result of the human detection system using the RD-03D microwave radar sensor connected to the Raspberry Pi Pico. The serial monitor displays messages such as “Human Detected” and “No Human Detected,” indicating the presence or absence of a person within the sensor’s range. When a person moves or stands near the sensor, it detects the motion or micro-movement and displays “Human Detected.” When there is no motion or presence, it shows “No Human Detected.” This output confirms that the system is working correctly, with proper communication between the sensor, microcontroller, and computer. It demonstrates that the project successfully detects human presence and can be used in automation or security applications.



Figure 5.2: Trajectory view

The output shown in the image represents the trajectory visualization of a microwave radar-based human presence detection system. The display is in the form of a polar radar map, where the green semicircular grid indicates the radar’s field of view. The radar sensor is positioned at the center bottom of the grid, and the radiating lines denote angular directions, typically ranging from -60° to $+60^\circ$. The concentric arcs represent distance intervals from the sensor, allowing easy estimation of how far a detected object is located. The colored dots seen within the radar field correspond to detected targets or moving objects. Each target’s position indicates its angle and distance relative to the radar sensor. On the left side of the display, parameters such as angle, distance, and speed are shown for each detected target. The angle represents the direction of the object from the radar’s centerline, while the distance indicates how far it is from the sensor. The speed values, measured in centimeters per second, reveal the object’s motion behavior—positive values mean the object is moving away from the radar, and negative values mean it is approaching. This visualization effectively shows real-time motion tracking, where the system identifies both stationary and moving targets, computes their trajectory, and displays their positions on the radar grid. Such an output is crucial during testing to validate the radar’s detection accuracy, measure object movement, and observe how well the system responds to human presence in its range.

5.2 Comparison with Existing Systems

The existing systems for motion detection and analysis primarily rely on traditional sensors such as ultrasonic, infrared (IR), and passive infrared (PIR) sensors. While these technologies are low-cost and easy to implement, they have several limitations, including low accuracy, restricted sensing range, and poor performance in adverse environmental conditions such as darkness, fog, or temperature variations. These sensors also fail to provide detailed information about the movement, such as speed, distance, and angular direction of the target.

In contrast, the proposed 24 GHz mmWave radar-based motion analysis system offers a significant improvement in terms of precision, robustness, and data richness. The new system uses the RD-03D mmWave radar module integrated with a Raspberry Pi Pico, allowing real-time detection of human presence and movement. Unlike the existing systems,

it provides continuous tracking with accurate distance, speed, and angle measurements even in low light or smoky environments. The processed radar data is visualized on a PC, enabling the generation of real-time trajectory maps that help in detailed motion analysis.

Overall, the proposed system overcomes the drawbacks of existing technologies by offering:

- Higher spatial resolution for accurate human detection and tracking.
- Reliable operation under diverse environmental conditions.
- Compact and cost-effective design suitable for security, healthcare, and automation applications.
- Enhanced data visualization through real-time trajectory plotting on PC interfaces.
- Thus, the 24 GHz mmWave radar integration with Raspberry Pi represents a modern and efficient alternative to conventional sensing systems.

5.3 Performance Evaluation

The performance of the proposed 24 GHz mmWave radar-based human motion detection system was analyzed based on several critical parameters such as accuracy, speed, reliability, range, and environmental adaptability. The system, which integrates the RD-03D mmWave radar sensor with a Raspberry Pi Pico microcontroller, was tested under different real-world conditions to validate its effectiveness and robustness.

The system demonstrated excellent results in all performance aspects, proving to be superior to conventional sensing systems like infrared (IR) or ultrasonic sensors. The following key observations summarize its performance:

Detection Accuracy: The radar achieved a high detection accuracy of around 95 percent, successfully identifying the presence and motion of human targets in both static and dynamic environments. It minimized false detections, even in cluttered indoor areas.

Distance and Speed Estimation: The radar provided precise distance and velocity measurements. The average range error was less than ± 5 cm, and the speed measurement error was within ± 0.1 m/s, allowing reliable short-range motion tracking.

Angular Detection: The system effectively determined the direction of movement, maintaining an angular accuracy of 3–5 degrees, which is sufficient for generating a clear motion trajectory.

Response Time and Real-Time Operation: The combination of the RD-03D radar and Raspberry Pi Pico ensured real-time processing with low latency. The radar data was processed instantly, and live trajectory visualization on the PC confirmed minimal delay between detection and display.

Environmental Stability: Unlike optical or IR-based systems, the mmWave radar operated efficiently under low-light, dusty, smoky, or obstructed conditions. This shows that the system is highly resistant to environmental variations.

System Reliability: The radar and processing unit maintained continuous, stable performance during extended operation, with no significant signal loss, communication delay, or overheating.

Power Efficiency: Operating on a 5 V power supply, the system consumed low power, making it ideal for portable, battery-powered, and IoT-based deployments.

Visualization Performance: The graphical interface displayed the detected trajectory, speed, and direction of movement in real time, improving the interpretability and usability of the data.

Overall, the performance evaluation confirms that the proposed mmWave radar motion analysis system delivers superior precision, robustness, and versatility compared to existing sensing technologies. It provides a cost-effective, compact, and energy-efficient solution for applications such as security monitoring, healthcare movement analysis, industrial automation, and smart home systems.

By combining high sensing accuracy, environmental resilience, and real-time data visualization, this system represents a significant advancement over traditional motion detection approaches.

Chapter 6

Conclusion and Future Enhancements

The system can be enhanced by integrating it with IoT cloud platforms like ThingSpeak, Blynk, or AWS IoT for remote monitoring and data visualization. This allows real-time transmission and storage of detection data, enabling users to view motion patterns and system performance from anywhere. Such integration also supports long-term data analysis and makes the system more suitable for smart home, healthcare, and automation applications.

6.1 Conclusion

The microwave radar-based human presence detection system successfully detects and tracks movement within its sensing range by analyzing distance, angle, and speed parameters. The radar visualization effectively represents the position and motion of detected targets in real time, confirming the system’s capability to monitor human activity accurately. The project demonstrates reliable performance in identifying motion direction and proximity without the need for optical or infrared sensors, making it suitable for various applications such as security systems, smart lighting, and automation. Overall, the system proves to be efficient, low-power, and adaptable for indoor environments, providing consistent detection even in low-light or obstructed conditions.

6.2 Limitations

- The detection accuracy can be affected by environmental factors such as obstacles, walls, and metallic reflections.
- The radar may generate false detections due to movement of non-human objects like fans or curtains.
- It cannot distinguish between multiple humans or objects when they are close to each other.
- The detection range and angular resolution are limited by the radar sensor’s hardware specifications.
- The system may require frequent calibration to maintain sensitivity and accuracy under varying conditions.
- External electromagnetic interference can impact the radar’s performance and signal stability.
- The system lacks visual identification, as it cannot provide images or recognize specific objects.
- Performance may decrease in complex indoor environments with many reflective surfaces.

6.3 Future Enhancements

- 1. Integration of machine learning or AI algorithms to differentiate between human motion and other moving objects.
- Improvement of range and angular resolution by using advanced radar modules or multi-antenna arrays.
- Addition of wireless communication features such as Wi-Fi, Bluetooth, or IoT connectivity for remote monitoring.
- Development of a mobile or web-based interface for real-time data visualization and control.

- Incorporation of environmental sensors (temperature, humidity, or light) for multi-parameter smart system applications.
- Implementation of data logging and cloud storage to analyze long-term human activity patterns.
- Enhancement of power efficiency through optimized signal processing and energy-saving modes.

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Appendix A

Figures/Tables/ Screen Shots

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Appendix B

Publications

[Publication Details/Certificate/Patents]